

Encoder Application — Social Media Shortage Detection

1. Introduction

This project develops an encoder-based natural language processing system to detect drug shortage-related discussions from Reddit. Drug shortages can affect patient access, treatment continuity, medication switching, and healthcare decision-making. However, early signals of shortage experience are often embedded in noisy social media posts, such as discussions about pharmacies being out of stock, refill delays, switching medications, or searching for alternative supply channels.

The goal of this project is to transform large-scale unstructured Reddit text into structured shortage-related signals. We focus on GLP-1 receptor agonist medications, including Ozempic, Wegovy, Rybelsus, Victoza, Mounjaro, and Zepbound. The project uses a transformer-based encoder model to classify whether a Reddit post is shortage-related, followed by additional filtering, topic modeling, and time-series aggregation. The resulting shortage discussion signals are then used as inputs for downstream analysis, including a VAR model to examine temporal relationships between social media discussions and prescription behavior, and a GNN-based prediction model to improve forecasting performance.

This report first describes the Reddit dataset and labeling process, then explains the NLP model, experimental setup, hyperparameter choices, classification results, and downstream use of the generated shortage signals.

2. Description of the Dataset

The dataset consists of approximately 800,000 Reddit posts and comments collected from 14 to 16 subreddits related to GLP-1 receptor agonists between January 2010 and January 2025. The

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sample covers multiple medications, including Ozempic, Wegovy, Rybelsus, Victoza, Mounjaro, and Zepbound, and spans several periods in which shortages became salient in public discussion. For supervised classification, two manually labeled datasets were used: 1000samples.xlsx and df_nokey_results.xlsx. We label them through 2 research assistants and ChatGPT, the overall kappa score is 95.6%. The main columns used in model training were the Reddit text, the binary shortage label, and the annotation reason.

The task was defined as a binary classification problem, where label 1 indicates shortage-related content and label 0 indicates non-shortage-related content. The labeled data were cleaned by standardizing label values such as “YES,” “NO,” “1,” and “0” into binary numerical labels. The full unlabeled Reddit corpus was then used for prediction after the final classifier was trained. In the full prediction stage, Reddit comments and submissions were merged, converted to lowercase, and classified in batches. The final prediction output was saved as predicted_results.csv, and the enriched result file was saved as updated_results.csv. For downstream time-series analysis, we converted Reddit timestamps into calendar dates and aggregated predicted shortage posts by day and week. Weekly aggregation was used to reduce high-frequency noise and to construct shortage discussion variables for later econometric and predictive modeling.

3. NLP Model and Algorithm

The core NLP model is a transformer-based encoder classifier. Specifically, we fine-tuned RoBERTa, a pretrained transformer encoder model, for binary sequence classification. The project README describes this component as an encoder-based NLP application designed to transform large-scale unstructured social media text into structured shortage signals.

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The model takes a Reddit post as input and outputs the probability that the post is shortage-related. Text is first tokenized using the RoBERTa tokenizer. Each post is converted into input IDs and attention masks, with truncation and padding applied to a fixed maximum sequence length. The transformer encoder then maps the token sequence into contextual embeddings, and a classification head predicts the binary label.

Formally, for each Reddit post x_i , the encoder produces a representation:

$$h_i = \text{Encoder}(x_i)$$

The classification layer maps this representation into logits:

$$z_i = Wh_i + b$$

The predicted probability of a shortage-related post is:

$$P(y_i = 1 | x_i) = \text{softmax}(z_i)_1$$

The model is trained using cross-entropy loss. Because shortage-related posts are relatively less frequent than non-shortage posts, the final version uses weighted cross-entropy:

$$L = - \sum_{c \in \{0,1\}} w_c y_c \log(\hat{y}_c)$$

where w_c is the class weight computed from the training label distribution.

In addition to classification, the pipeline includes topic modeling using LDA. After predicted shortage posts were identified, the text was tokenized, stop words were removed, and words

were lemmatized. A 10-topic LDA model was then used to explore the major themes in shortage-related discussions, such as pharmacy access, switching behavior, dosage issues, and availability concerns. The README confirms that the full pipeline includes model training, full corpus prediction, topic modeling, time-series analysis, and drug-specific analysis.

Figure 4 summarizes the overall pipeline. We first collected Reddit posts, labeled a subset of the data, trained an encoder-based shortage classifier, applied the model to the full corpus, filtered shortage-related discussions, and finally aggregated the predicted signals into weekly time-series variables for downstream analysis.

4. Experimental Setup

The classification task was implemented using Python, PyTorch, Hugging Face Transformers, scikit-learn, pandas, gensim, nltk, and spaCy. The main training script was `shortage_identification.py`, which includes data loading, cleaning, augmentation, model training, prediction, topic modeling, and time-series analysis. The labeled data were split into training, validation, and test sets using a 60/20/20 split. The training set was used to optimize model parameters, the validation set was used for model selection and early stopping, and the test set was used for final performance evaluation.

The input text was tokenized with `AutoTokenizer.from_pretrained("roberta-base")`. Two maximum sequence lengths were considered: 128 tokens for the baseline model and 512 tokens for the improved model. The baseline model used standard cross-entropy loss without class weights. The final model introduced several improvements: class weighting, weighted loss, partial layer freezing, longer sequence length, and early stopping. To evaluate performance, we used accuracy, precision, recall, and F1 score. F1 score was treated as the main metric because

the task involves class imbalance and because both false positives and false negatives are important. False positives may incorrectly identify general medication discussions as shortage-related, while false negatives may miss important shortage signals.

After training, the final model was applied to the full Reddit corpus. Predictions were generated in batches, and each post was assigned a binary predicted label. The predicted shortage posts were then aggregated into daily and weekly counts. Drug-specific time series were also constructed using keyword matching for medications such as Wegovy, Ozempic, Mounjaro, Zepbound, Tirzepatide, and semaglutide.

5. Hyperparameter Search and Overfitting Prevention

We searched and adjusted several key hyperparameters, including learning rate, batch size, number of epochs, maximum sequence length, weight decay, warmup steps, and classification threshold. The main hyperparameter settings were:

<i>Hyperparameter</i>	<i>Value</i>
<i>Model</i>	RoBERTa-base
<i>Batch size</i>	16
<i>Epochs</i>	20
<i>Learning rate</i>	2e-5 / 3e-5
<i>Max sequence length</i>	128 / 512
<i>Weight decay</i>	0.01 / 0.02
<i>Warmup steps</i>	500
<i>Early stopping patience</i>	3

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The baseline model used learning rate $2e-5$, batch size 16, max length 128, and weight decay 0.01. The improved model used learning rate $3e-5$, max length 512, weight decay 0.02, warmup steps of 500, and early stopping. Several strategies were used to reduce overfitting. First, the labeled data were separated into training, validation, and test sets, so model performance was evaluated on unseen data. Second, early stopping was used to stop training when validation F1 no longer improved. Third, weight decay was applied as regularization. Fourth, the embedding layer and the first eight RoBERTa encoder layers were frozen in the improved model, reducing the number of trainable parameters and limiting overfitting on the relatively small labeled dataset. Finally, we used word-swap data augmentation on target words such as “shortage,” “short,” and “available” to reduce keyword leakage and encourage the model to learn broader semantic patterns rather than relying only on obvious keywords. The README specifically identifies word swapping as a data augmentation step designed to reduce keyword leakage and improve generalization.

6. Results

The final encoder model achieved strong classification performance on the validation and test data. In the manually annotated sample, 1,200 posts were labeled, with 421 identified as shortage-related. Annotation consistency was high, with Cohen’s Kappa of 0.85, indicating reliable agreement in the labeling process.

The fine-tuned transformer model achieved an F1 score of 0.87 for the positive shortage-related class and 0.94 for the negative non-shortage class on the validation set. This indicates that the model performed well in distinguishing shortage-related discussions from general GLP-1 medication discussions.

After applying the trained classifier to the full Reddit corpus, the model identified approximately 88,554 potentially shortage-related posts. We then applied a second-stage LLM-based filtering process using multi-layer prompting with an o3-mini model to further restrict the corpus to posts specifically discussing drug shortages and coping strategies. A manual inspection of 200 filtered posts showed positive class accuracy of 0.98 and negative class accuracy of 0.94. The final shortage corpus contained approximately 52,866 posts.

The time-series analysis showed that shortage-related Reddit discussions were not evenly distributed over time. Instead, they appeared in waves that corresponded to periods of public concern about GLP-1 medication availability. Weekly aggregation reduced noise and made the trend more suitable for downstream modeling. Drug-specific plots further showed that shortage discussions varied across medications, with Wegovy and Ozempic receiving particularly strong attention during major shortage periods.

Topic modeling provided an exploratory view of the content of shortage-related discussions. The LDA topics captured recurring themes such as pharmacy search behavior, refill problems, medication switching, dosage concerns, and user strategies for dealing with unavailable drugs. These results suggest that Reddit discussions contain not only general complaints, but also behavioral signals about how patients respond to drug shortages.

Overall, the results show that an encoder-based NLP model can effectively convert noisy Reddit text into structured shortage signals. These signals provide a useful bridge between unstructured social media discussions and quantitative downstream models.

7. Summary and Conclusions

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This project developed a transformer-based NLP pipeline to detect drug shortage-related discussions from Reddit. Starting from a large corpus of approximately 800,000 posts and comments, we constructed a labeled training set, fine-tuned a RoBERTa-based classifier, applied the model to the full corpus, and generated structured shortage discussion signals. The final model achieved strong classification performance, with high F1 scores for both shortage-related and non-shortage-related classes.

The main lesson from this project is that social media data can be transformed into meaningful behavioral signals when combined with supervised NLP classification and careful validation. Reddit posts are noisy and informal, but they contain valuable real-time information about patient experiences, medication availability, and coping strategies during shortages.

The output of this NLP model is not only a classification result. It also supports broader business and healthcare analytics. In our downstream analysis, the weekly shortage discussion signals are used in a VAR model to examine temporal relationships and potential causality between social media discussions and prescription behavior. In addition, these NLP-derived features improve the predictive capacity of a GNN-based forecasting model, suggesting that social media contains useful information beyond traditional prescription and event data.

Future improvements could include expanding the labeled dataset, testing additional transformer models, improving the second-stage filtering process, using more advanced topic models such as BERTopic, and integrating explainability methods such as SHAP or integrated gradients to better understand which linguistic features drive shortage classification.